Universidade do Minho
Departamento de Informática

Dados e Aprendizagem Automática
Ano letivo 2026/2027

Practical Exercise no. 7

Theme

Interpretability and Explainability

Exercise

While **interpretability** focuses on understanding the decision-making process of a model, **explainability** focuses on communicating this process to the user. Both are important because they provide insight into how machine learning algorithms make decisions and are essential to ensure that the models are reliable, secure, and adhere to the appropriate ethical principles.

Tasks

This practical exercise includes the following tasks:

T1. Follow and implement the steps presented in the lab notes to achieve local and global interpretability of features and models.

T2. Analyze feature importance of the Titanic dataset. Which three features have the highest decision weight? Use Random Forest Feature Importance, Mean Decrease in Impurity and Permutation Importance to analyze this.

T3. Analyze and compare the results obtained for the Random Forest classifier and the Linear Regression model using SHAP values. Which three features were more relevant for predicting passenger survival in each model? How can the model's decision be explained?

T4. Conduct an analogue study using XGBoost. Compare these results with those from the previous models. Which features are the most important for determining the target variable? Which model is more interpretable/explainable?